

01 — Product Documentation

Complete documentation suite (Markdown → PDF with Mermaid diagrams)

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Diagrams rendered with Mermaid 11

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Product Overview

Status labels: Content below is **Verified** from the monorepo unless marked otherwise.

Product name

OPMS — Order / Operations Portal Management System (packages also use branding names such as Medica / Shakti in env examples and package names).

Requires confirmation: Official marketed product name and legal entity branding for end customers.

Sources:

- DOCKER.md
- opms-backend/package.json (opms-backend)
- auth-service/package.json (medica-auth-service)
- .env.docker.example (COMPANY_NAME / NEXT_PUBLIC_COMPANY_*)

Purpose

OPMS is a multi-portal business operations platform that enables:

1. **Order lifecycle management** — create, approve, finance/account review, dispatch, transport, delivery, returns, due sheets, flags, reminders
2. **Master data** — parties (customers/suppliers), products (including kits), zones, fleet (vehicles, drivers, transport agents)
3. **CRM** — Lead Manager (leads, follow-ups, quotations, terms)
4. **Field operations** — Work Planner (daily plans, visits, works, expenses)
5. **Identity & access** — Auth service + User Manager (users, departments, roles, portals, company info)
6. **Communications** — in-app notifications, web push, email, WhatsApp messaging

Problem being solved

Inferred from implementation: Organizations need a single identity layer and coordinated portals for sales-to-dispatch order operations, lead-to-quote CRM, and field sales planning — instead of disconnected tools.

Requires confirmation: Exact business problem statement and ROI goals from product/business team.

Target users

Portal / App	Typical users (from code)
App Frontend (SSO hub)	Any authenticated user with department + roles
OPMS	super_admin , admin , sales , finance , account , dispatch
User Manager	Super-admin / admin style access (hasSuperAdminAccess)

Portal / App	Typical users (from code)
Lead Manager	admin , manager , executive
Work Planner	admin , manager , executive (+ global admin bypass)

Sources:

- `opms-backend/src/middlewares/opmsAuth.middleware.js`
- `lead-manager-backend/src/middlewares/leadManagerAuth.middleware.js`
- `work-planner-backend` work planner auth middleware
- Frontends' `authStorage` / middleware

Main use cases

1. Log in via App Frontend and launch assigned portals (SSO with `?token=`).
2. Create and progress sales orders through departmental approvals to dispatch/delivery.
3. Maintain party and product masters (and party-product rates).
4. Plan transport and manage fleet/agents.
5. Manage leads, follow-ups, and quotations; convert leads.
6. Plan field work, check in/out visits, submit expenses.
7. Administer users, portal assignments, departments/roles, and company branding.
8. Receive notifications (SSE/in-app/web push) and send transactional messages (email/WhatsApp).

Major capabilities

- JWT authentication with shared `JWT_SECRET`
- Portal-scoped RBAC (`user.portals[].portal_code` + `access_roles`)
- Soft-delete / restore on many domain entities
- BullMQ background jobs (orders, notifications, messages) on Redis
- External file management API for uploads
- Google Sheets webhook ingest (orders, parties, products, reminders)
- Dual-tenant host layout (spspl / medica nginx configs)

Product boundaries

In scope (verified in repo)

- Microservices and micro-frontends listed in `docker-compose.yml`
- Order workflow as implemented in `workflow.transitions.js`
- Lead Manager and Work Planner modules
- Auth, company info, portals, departments
- Messaging and notifications services

Out of scope / not identified in codebase

- Payment gateway / online payment capture — **Not identified** as implemented (payment_status fields exist; collection/paid lifecycle in essentials docs is **stale**)
- Native mobile apps — **Not identified**
- Embedded analytics BI warehouse — **Not identified** (dashboards are API summaries only)

- In-compose MongoDB or MinIO — **external** to Compose
- Automated CI/CD pipelines — **Not identified** (no `.github/workflows`)

Related docs

- [PRD](#)
- [Scope](#)
- [Features](#)
- [System Architecture](#)

Product Requirements Document (PRD)

Source of truth: Implementation in this repository. Numerical SLAs are **not** defined in code unless noted.

Product Vision

Provide a portalized operations suite where a company can authenticate once, assign users to OPMS / Lead Manager / Work Planner / User Manager portals, and execute order-to-delivery, lead-to-quotation, and field-work processes with shared masters and communications.

Requires confirmation: Formal vision statement from product leadership.

Problem Statement

Sales, finance, account, and dispatch teams need coordinated order processing with auditability; sales leadership needs lead pipelines and quotations; field teams need daily work plans and expense capture — under one identity and company branding model.

Objectives

ID	Objective	Evidence
O-1	Support multi-role order workflow with transitions and history	<code>workflow.transitions.js</code> , OrderWorkflow
O-2	Centralize identity and portal access	auth-service User/Portal
O-3	Provide CRM for leads and quotations	lead-manager-*
O-4	Support field work planning and expenses	work-planner-*
O-5	Notify users and send communications	notification-service, message-service
O-6	Deploy as Docker Compose + nginx	<code>docker-compose.yml</code> , <code>nginx/</code>

Target Users

See Personas and Roles.

User Personas

P1 — Sales Executive (OPMS + optional Lead/Work Planner)

- Creates orders, views own pipeline, may manage leads/plans depending on portal assignment.

P2 — Finance / Account Reviewer

- Reviews approvals, due sheets, rates, finance/account queues.

P3 — Dispatch / Logistics

- Creates dispatches, manages transport planner, fleet, deliveries.

P4 — Portal Administrator (User Manager)

- Manages users, portals, departments/roles, company info.

P5 — Lead Manager / Manager

- Oversees team leads, quotations, terms, reports.

P6 — Work Planner Manager

- Assigns team settings, approves plans/expenses.

User Stories

As a sales user,
I want to create and submit an order,
so that finance/account/dispatch can process it.

As a finance user,
I want to approve or reject order approvals,
so that commercial risk is controlled before dispatch.

As a dispatch user,
I want to create dispatches and transport plans,
so that goods can be shipped and tracked.

As any authenticated user,
I want to log into App Frontend and open my assigned portals with SSO,
so that I do not re-authenticate for each app.

As a super admin,
I want to assign portal access_roles to users,
so that each person only sees authorized workspaces.

As a lead executive,
I want to create leads, schedule follow-ups, and convert won leads,
so that the sales pipeline is managed in one place.

As a work planner executive,
I want to create a daily plan with visits and submit expenses,
so that field activity is recorded and approved.

Functional Requirements

ID	Requirement	Source
FR-001	Users can authenticate with email/password and receive JWT	auth-service POST /api/auth/login

ID	Requirement	Source
FR-002	JWT can be exchanged across frontends via <code>?token= SSO</code>	app-frontend / portal AuthTokenSsoBootstrap
FR-003	OPMS domain APIs require <code>portal_code opms + access_roles</code>	opmsAuth.middleware.js
FR-004	Orders support create, submit, transition, soft-delete, restore	order.routes.js
FR-005	Order status transitions follow allowed graph	workflow.transitions.js
FR-006	Order approvals support approve/reject/send-to-finance/account/amend	orderApproval.routes.js
FR-007	Dispatches, deliveries, returns, due sheets, unbilled orders are manageable	respective routes
FR-008	Transport planner supports plan lifecycle and order line actions	transportPlanner.routes.js
FR-009	Parties and products CRUD (+ soft-delete) via party/product services	party-service, product-service
FR-010	Lead CRUD, follow-ups, convert, quotations, terms	lead-manager-backend
FR-011	Work plans, visits, works, expenses with approve/reject	work-planner-backend
FR-012	Notifications list/stream/read + web push subscribe	notification-service
FR-013	Email/WhatsApp send and communication triggers	message-service
FR-014	Attachments upload via file management API	attachments modules
FR-015	Dashboards per OPMS role	dashboard.routes.js
FR-016	Google Sheet webhooks for orders/parties/products/reminders	webhook routes + secret
FR-017	Company branding via CompanyInfo	auth-service company-info
FR-018	Soft-delete with trash list and restore on many entities	softDelete plugin

Non-Functional Requirements

See [Non-Functional Requirements](#). No latency/availability SLAs found in repository.

Acceptance Criteria (implementation-derived)

Story area	Acceptance criteria (verified behavior)
Login	Valid credentials return JWT; <code>/api/auth/me</code> returns user with portals
OPMS access	User without <code>opms</code> portal receives 403 on domain routes
Order transition	Invalid next status rejected per transitions graph
Soft-delete	Soft-deleted records excluded from default lists; restore returns them
SSO	Opening portal URL with <code>?token=</code> establishes session and strips token
Health	GET <code>/health</code> on backends returns success when process is up

Requires confirmation: Formal UAT scripts and business acceptance sign-off.

Scope

In scope (verified)

Applications

Area	Components
Frontends	app-frontend, opms-frontend, user-manager-frontend, lead-manager-frontend, work-planner-frontend
Backends	opms-backend, auth-service, product-service, party-service, lead-manager-backend, work-planner-backend, message-service, notification-service
Infra in Compose	Redis
Edge	Host nginx configs under <code>nginx/</code> for spspl and medica hostnames
Data store	MongoDB (external URI)
Files	External File Management API (<code>FILE_MANAGEMENT_API_*</code>)

Functional domains

- Authentication, users, departments, roles, portals, company info
- Orders and departmental workflow
- Approvals, finance queues, account steps
- Dispatch, transport shipments, deliveries, returns
- Due sheets, unbilled orders, final order statements
- Flags, reminders, attachments, activity logs
- Fleet: vehicles, drivers, transport agents
- Transport planner
- Party/zone/product catalogs and party-product rates
- Leads, follow-ups, quotations, lead masters, terms
- Work plans, visits, works, expenses, user work-planner settings
- Notifications, web push, email, WhatsApp, auto-emails, communication helpers
- Google Sheets webhook ingestion (selected modules)

Out of scope / not implemented in repo

Item	Status
Compose-hosted MongoDB	Not in compose — external
Compose-hosted MinIO	Leftover <code>minio-data/</code> only; not a service
RabbitMQ	Not present — BullMQ/Redis used
CI/CD pipelines	Not identified
Mobile native apps	Not identified

Item	Status
Full payment collection / invoice_generated lifecycle in live transitions	Essentials docs stale; not in <code>workflow.transitions.js</code>
Automated Mongo backup jobs in repo	Not identified
Centralized APM/metrics stack	Not identified

Partial / dual-tenant

- `docker-compose.prod.yml` defines a **subset** of services — incomplete standalone full deploy (**Verified**).
- Medica tenant uses alternate ports/nginx (5001–5013 / `*.medicaent.in`) via `.env.docker.medica` (**Verified** configs exist; dual-stack runbook **Requires confirmation**).

Related

- [Features](#)
- [Integrations](#)

Personas and Roles

Identity model (verified)

Users are stored in **auth-service** with:

- `department` (string; seeded values include `super_admin` , `admin` , `sales` , `finance` , `account` , `dispatch`)
- `roles[]` → Role documents (`sales_executive` , `finance_manager` , etc.)
- `portals[{ portal, portal_code, access_roles[] }]`

Sources: `auth-service/src/models/User.js` , `Role/Department/Portal` models, seed services.

Portals

portal_code	Seeded in DEFAULT_PORTALS?	access_roles (observed)
opms	Yes	<code>super_admin</code> , <code>admin</code> , <code>sales</code> , <code>finance</code> , <code>account</code> , <code>dispatch</code>
work_planner	No (used in code)	<code>admin</code> , <code>manager</code> , <code>executive</code> (+ <code>super_admin</code> /sales observed in constants)
lead_manager	No (used in code)	<code>admin</code> , <code>manager</code> , <code>executive</code>
user_manager	No (frontend gate)	Unknown exact role list — frontend checks <code>admin</code> / <code>super_admin</code> style access

Requires confirmation: Official portal catalog and which portals must be seeded for production onboarding.

OPMS access roles → UI workspaces

access_role	Frontend portal path	Typical focus (inferred from nav)
<code>super_admin</code>	<code>/super_admin</code>	Full masters + User Manager embed
<code>admin</code>	<code>/admin</code>	Broad order/masters/transport
<code>sales</code>	<code>/sales</code>	Orders overview/create (own data emphasis)
<code>finance</code>	<code>/finance</code>	Finance review, masters, transport
<code>account</code>	<code>/account</code>	Account review, masters, transport
<code>dispatch</code>	<code>/dispatch</code>	Dispatch, transport planner/agents

Source: `opms-frontend` portal nav and middleware.

Lead Manager roles

Role	Capabilities (from UI/API gates)
admin	Full leads; quotations; terms; team reports
manager	Similar to admin for quotations/terms; team oversight
executive	Leads/follow-ups/personal reports; no quotations nav

Work Planner roles

Role	Capabilities
admin	Team manager settings; approve plans/expenses
manager	My team settings; approvals
executive	Own plans, visits, expenses

App Frontend access

Requires JWT with **department** and at least one **role**; sidebar built from `role_codes` ; portal tiles from `portals[]` .

Persona mapping

See PRD personas P1–P6 in 02-prd.md.

Features

Feature catalog derived from routes and frontend screens. **Verified** unless noted.

A. App Frontend (SSO Hub)

Feature	Description	Source
Login	Email/password login with company branding	app-frontend /
Department dashboard	Assigned portals launcher	/dashboard/[department]
Profile / password	Change password	/dashboard/profile
Notifications	Bell via auth proxy	NotificationBell
SSO launch	Opens peer apps with ?token=	PortalsSection

B. OPMS Frontend / Backend

Feature	Description
Role workspaces	admin/sales/finance/dispatch/account/super_admin
Order master	List by workflow/priority; create; detail with lifecycle tabs
Order workflow	Submit, transition, history, assignees, fulfillment
Approvals	Unified OrderApproval actions
Finance queues	/api/finance/queue , /summary
Dispatch	Create/patch soft-delete with bill document upload
Deliveries / returns	Log shipment delivery; returns CRUD
Due sheets / unbilled / final statements	Billing support artifacts
Flags	Raise/manage order flags (blocking types in workflow rules)
Reminders	Reminders + follow-ups; Google Sheet webhook
Parties / zones	Via party-service proxy
Products / groups / brands / manufacturers / kits	Via product-service proxy
Fleet	Vehicles, drivers, transport agents
Transport planner	Plans, calendar, LR, packed/dispatched/delivered
Attachments / files	Upload + view/download redirects
Dashboards	Per-role KPI endpoints
Activity log	List activity
Terms & conditions	OPMS T&C API (note: router lacks requireAuth — see security)

Feature	Description
Communication	Proxied to message-service
Push notifications	Proxied to notification-service
PDF/export helpers	Frontend html2canvas/jspdf

C. User Manager

Feature	Description
User directory CRUD	List/add/edit/detail
Portals registry	Seed defaults; assign per user
Departments & roles	Hub management
Company information	Branding and legal/tax fields
KPI cards	User counts

D. Lead Manager

Feature	Description
Lead pipeline	CRUD, assign, status, qualify, mark lost, convert
Follow-ups	Calendar + reminders (cron digests)
Reports	Admin “Sales Reports” / others “My Reports”
Quotations	CRUD, submit/approve/reject, PDF, email
Terms & conditions	Master for quotations
Bulk / Google Sheet ingest	Lead import paths
Attachments	Lead attachments module

E. Work Planner

Feature	Description
Work plans	Create/list/calendar; submit/approve/reject/complete
Visits / works	Nested lifecycle (check-in/out/complete)
Expenses	Submit/approve/reject with receipts
Assigned users / settings	Manager templates, CC emails, plan types
Day-end / exports	PDF/Excel helpers in UI

F. Platform services

Feature	Description
Auth	Login, me, change-password
Notifications	Internal create, SSE stream, mark read, web push
Messages	WhatsApp webhook, send message/email, auto-email trigger, communication helpers

Business Requirements

Mapped from implemented behavior. Formal BR IDs from business analysts: **Requires confirmation.**

BR ID	Statement	Implementation evidence
BR-001	System shall authenticate users and issue time-bounded tokens	JWT + JWT_EXPIRES_IN
BR-002	Access shall be portal-scoped	portals[].portal_code + middleware
BR-003	Orders shall follow controlled status transitions	workflow.transitions.js
BR-004	Certain flags shall block dispatch/transit/delivery transitions	workflow.rules.js flag blocks
BR-005	Pricing shall lock after finance/account/dispatch stages	pricing lock statuses in rules
BR-006	Soft-deleted records shall be recoverable by authorized users	softDelete plugin + restore routes
BR-007	Company branding shall be configurable	CompanyInfo module
BR-008	Field teams shall plan daily work and claim expenses	Work Planner models/APIs
BR-009	Sales shall manage leads through quotation to conversion	Lead Manager
BR-010	System shall notify users of relevant events	notification-service + queues
BR-011	System may send email/WhatsApp for order/comms	message-service integrations
BR-012	External sheets may ingest masters/orders via webhook + secret	Google Sheet webhook routes

Priority

Not identified in repository (no MoSCoW tags in code).

Related

- [Business Rules](#)
- [PRD](#)

Non-Functional Requirements

Only document NFRs evidenced in configuration/code. Do not invent numeric SLAs.

Security

NFR	Status	Evidence
JWT authentication	Implemented	auth-service, shared JWT_SECRET
Portal RBAC	Implemented	middleware per portal
Password hashing	Implemented	bcryptjs
HTTPS at edge	Configured for prod hosts	nginx ssl snippets
Secrets via env	Documented	<code>.env.docker</code> , DOCKER.md
Break-glass master password	Optional env	MASTER_PASSWORD — leave empty in prod (DOCKER.md)
Rate limiting	Not identified as middleware	—
Redis AUTH	Not configured in compose	security risk

Performance

NFR	Status
JSON body limit configurable	JSON_BODY_LIMIT (example 50mb)
Nginx proxy timeouts	API hosts use extended timeouts (e.g. 120s) and large body (50m) on api-opms
Explicit p95 latency targets	Not identified

Availability / Scalability

NFR	Status
Container restart policy	<code>restart: unless-stopped</code>
Horizontal scaling playbooks	Not identified
Redis health dependency for queues	message/notification/opms-backend

Reliability

NFR	Status
Health endpoints	<code>/health</code> on backends
BullMQ retry/attempts	Message model tracks attempts; exact policies Requires confirmation from queue configs
Soft-delete over hard-delete	Many domain models

Maintainability

NFR	Status
Modular Express modules	<code>src/modules/*</code>
OpenAPI for opms-backend	<code>/api-docs</code>
Shared schema registries duplicated across services	Maintainability risk — documented in audit

Auditability

NFR	Status
OrderWorkflow / OrderStatusHistory	Append-only workflow audit
ActivityLog	Present
Who approved (signatures on OrderApproval)	Implemented

Accessibility / i18n

Not identified as systematic requirements in frontend code.

Related

- [Security](#)
- [Performance](#)

Business Rules

Order status transitions (verified)

Source: `opms-backend/src/modules/workflow/workflow.transitions.js`

```
draft → submitted, cancelled
submitted → sales_approved, finance_review, finance_rejected, on_hold, cancelled
sales_approved → finance_review, finance_rejected, on_hold, cancelled
finance_review → finance_approved, finance_rejected, on_hold, dispatch, cancelled
finance_rejected → submitted, cancelled
finance_approved → account_review, dispatch, on_hold, finance_rejected, cancelled
account_review → account_approved, account_rejected, finance_rejected, on_hold, cancelled
account_rejected → account_review, finance_approved, cancelled
account_approved → dispatch, account_review, on_hold, finance_rejected, account_rejected,
cancelled
dispatch → in_transit, on_hold, finance_rejected, cancelled
in_transit → delivered, on_hold, cancelled
delivered → in_transit, cancelled
on_hold → (resume to multiple prior active statuses), cancelled
cancelled → (none)
```

Note: Schema includes `closed` but transition graph has **no** `closed` edges. `POST .../close-after-full-delivery` exists — behavior should be verified against service code when documenting close semantics.

Department permission to set status

Source: `workflow.rules.js` (`DEPT_CAN_SET_STATUS` and special cases).

Sales is limited; admin/super_admin broader; finance/account/dispatch scoped.

Flag blocking

Flag types `payment_issue`, `stock_issue`, `dispatch_issue` **block** transitions into `dispatch` / `in_transit` / `delivered`.

Pricing lock

Pricing locked after statuses including `finance_approved`, `account_approved`, `dispatch`, `in_transit`, `delivered` (per rules).

Priority derivation

Order `priority` derived from expected delivery date windows (>10 low, 5–10 normal, 3–4 high, ≤2 urgent) — as documented in schema analysis of Order model.

Soft-delete

Default queries exclude `deletedAt != null`. Restore endpoints reverse soft-delete.

Portal access

Missing portal assignment ⇒ 403 on portal-gated APIs even if `department` is set (swagger description + middleware).

Lead / Work Planner state machines

Enums exist on models; full transition matrices beyond enums: **partially documented** in functional docs — deep service validation **Requires confirmation** for edge cases.

Stale documentation warning

`opms-backend/docs/essentials/*` describes older payment/invoice/collection flows — **do not** treat as live business rules.

Integrations

Integration	Purpose	Config / evidence	Status
MongoDB Atlas / external Mongo	Primary datastore	MONGO_URI / MONGODB_URI	Required
Redis	BullMQ queues + cache backbone	REDIS_URL , compose redis	Required for queue services
File Management API	Uploads / MinIO-backed files	FILE_MANAGEMENT_API_URL , FILE_MANAGEMENT_API_KEY	Optional
SMTP	Email fallback	SMTP_*	Optional
Microsoft Graph	Preferred outbound email	MICROSOFT_GRAPH_*	Optional
WhatsApp Cloud API	Templates + webhooks	WHATSAPP_*	Optional
Web Push (VAPID)	Browser push	VAPID_*	Optional
Google Sheets webhooks	Ingest orders/parties/products/reminders	GOOGLE_SHEET_WEBHOOK_SECRET	Optional
Host nginx + Let's Encrypt	TLS termination	nginx/	Prod edge
Inter-service HTTP	BFF proxies	*_SERVICE_URL	Required in compose

Not identified

- Payment gateways (Razorpay/Stripe/etc.)
- SMS providers beyond WhatsApp
- External ERP connectors (SAP/etc.)
- RabbitMQ / Kafka

Related

- [Integration Architecture](#)

Product Glossary

Term	Meaning in this codebase
OPMS	Core order operations portal and its backend BFF
Portal	Application workspace assigned on a user (<code>portal_code</code>)
access_roles	Role strings within a portal (e.g. <code>finance</code> under <code>opms</code>)
Party	Customer/supplier (or both) master record
Order	Central commercial/fulfillment document with workflow status
OrderApproval	Unified approval document for admin/finance/account signatures
Due sheet	Documented due/commercial sheet attached to an order
Unbilled order	Tracking record for unbilled quantities/status
Dispatch	Warehouse dispatch against an order
Transport plan	Planned multi-order shipment plan for an agent/date
Lead	CRM prospect in Lead Manager
Quotation	Lead quotation with approval/status enums
Work plan	Daily field plan for a sales user
Soft-delete	Record marked with <code>deletedAt</code> rather than removed
SSO hub	app-frontend launching other apps with JWT query token
BFF	Backend-for-frontend pattern — <code>opms-backend</code> proxies sibling APIs
BullMQ	Redis-backed job queue library
CompanyInfo	Company branding and legal/tax profile

Related

- [Features](#)
- [Database Overview](#)